Econ 201B: Lecture 2

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1 Proof of Domination Theorem Cont.

Proof. Recall from the previous lecture that we can use the SHT between the set payoffs generated by mixed strategies and the set of undominated actions to back out a vector $p > 0$ s.t.

$$p \cdot \vec{u}(\sigma) \leq \alpha \leq p \cdot x$$

$\forall \sigma \in \Delta(A)$ and $\forall x \in \mathcal{C} = \left\{ z \in \mathbb{R}^{\Omega} : z >> \vec{u}(\sigma^*) \right\}$. We can then define $\mu \in \mathbb{R}^{\Omega}$ by

$$\mu(\omega) = \frac{p(\omega)}{\sum_{\tilde{\omega} \in \Omega} p(\tilde{\omega})}$$

and so $\mu(\omega) \geq 0 \ \forall \omega$ and $\sum_{\omega \in \Omega} \mu(\omega) = 1$ is a probability and $\mu \in \Delta(\Omega)$.

Also recall that if an action is undominated, then it is optimal for some beliefs. Because μ is obtained from p by multiplication of a positive scalar, then

$$\mu \cdot \vec{u}(\sigma) \leq \mu \cdot x, \forall \sigma \in \Delta(A), \forall x >> \vec{u}(\sigma^*)$$

Now notice that $\forall n > 0$, this means

$$\mu \cdot \vec{u}(\sigma) \leq \mu \left(\vec{v}(\sigma^*) + \left(\frac{1}{n}, ..., \frac{1}{n} \right) \right) \to \mu \cdot \vec{\mu}(\sigma^*)$$

so, we can conclude that $\forall \sigma \in \Delta(A)$,

$$\begin{gathered} \mu \cdot \vec{u}(\sigma) \leq \mu \cdot \vec{u}(\sigma^*) \\ \Leftrightarrow \\ \mathbb{E}_{\mu}[u(\sigma, \tilde{\omega})] \leq \mathbb{E}_{\mu}[\mu(\sigma^*, \tilde{\omega})] \end{gathered}$$

□

2 Perfect Information Games "PIGS"

Let's consider the simplest form of games where agents have perfect information.

2.1 Definitions and notation

Let A be a set and a function mapping $[n] = \{1, ..., n\} \to A$ is a sequence of length n in A. We can write

$$h = (a_1, ..., a_n)$$

to be a sequence that contains information up to a_n. Let A^n denote the set of all sequences of length n and $\cup_{n \in \mathbb{N}} A^n$ denote all finite sequences in A.

When

$$h' = \left(a'_1, ..., a'_n, a'_{n+1}, ...\right)$$

is a (finite or infinite) sequence, and $h = (a_1, ..., a_n)$ is a sequence with $a'_i = a_i$ then we say that h' is a continuation of h and we say that h is a prefix of h'.

The elements of A are called actions and the sequences are called histories. We will have a set H of histories (where $h \in H$) as primitive. These are taken as given. By convention, we will always have H containing a length-zero history $\emptyset$ call the null history. A history h is a sub-history h' if it is a prefix of h'. We also have that a history H is terminal if it has no continuation in H and we denote the set of terminal histories as Z.

2.2 PIG

Definition 1. An extensive form game with perfection information (a PIG) G is a tuple

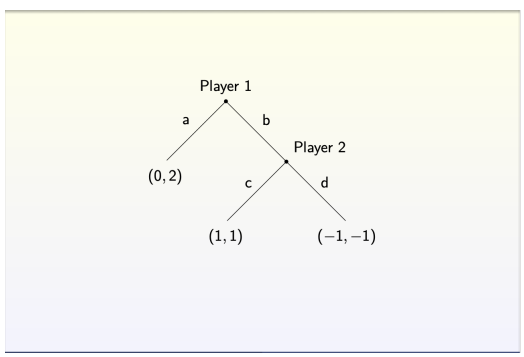
$$G = \left(N, A, H, P, \{u_i\}_{i \in \mathbb{N}}\right)$$

where

1. N is the set of players
2. A is the set of actions
3. H is the set of histories such that
   (a) If $h \in H$ then every prefix of h is also in H.
   (b) Sequences in H need not be finite. If they are not, then we need that if H contains every prefix some infinite sequence h then it also contains h.
   (c) Any infinite history in H is terminal.
   (d) Z is the set of terminal history in H.
4. P is a function from $H \backslash Z$ to N.
5. For each player $i \in N$, $u_i : Z \to \mathbb{R}$ is a utility (or payoff) function

A PIG is a called a **finite horizon game** if there is a bound on the length of histories in H. A PIG is called **finite** if N and H are finite.

Consider the following example where the PIG is written as a tree,



We can also rewrite formally as

$$\begin{aligned} N &= \{1, 2\} \\ A &= \{a, b, c, d\} \\ H &= \{\emptyset, (a), (b), (b, c), (b, d)\} \\ Z &= \{(a), (b, c), (b, d)\} \\ P(\emptyset) &= 1, P(b) = 2 \end{aligned}$$

This reasoning can be extended for larger trees.

3 Strategies

We can also think of how players respond at different points in the game.

3.1 Set up

Definition 2. First, let's denote by $A(h)$ to be the actions available to player $P(h)$ after history h. In this case, h is the prefix up to that point. Formally,

$$A(h) = \{a \in A : ha \in H\}$$

Definition 3. For each player $i \in N$, we can also define the set of histories at which i movies. Let

$$H_i = \{h \in H \backslash Z : i = P(h)\}$$

Definition 4. A strategy for player i is a function $s_i : H_i \to A$ for which

$$s_i(h) \in A(h), \forall h \in H_i$$

and let S_i be the set of all strategies available to player i.We can think of a strategy as a complete contingent plan for playing the game.

Definition 5. A strategy profile is a collection of strategies for each of the players,

$$s = (s_i)_{i \in N}$$

Given a strategy profile s in game G, we have a terminal history $h(s) \in Z$. We can then write payoffs in terms of strategy profiles

$$u_i(s) = u_i(h(s))$$

Consider again the simple example from earlier. The strategy for player 1 is given by

$$S_1 = \{a, b\}$$

and the strategy for player 2 is given by

$$S_2 = \{c, d\}$$

3.2 The Ultimatum Game

Let's apply the notations and definitions to the ultimatum game. In this game, player 1 makes an offer $a \in \{0, 1, 2, 3, 4\}$ to player 2. Player 2 either accepts or rejects. If player 2 accepts than she receives a dollars and player 1 receives $4 - a$ dollars. If player 2 rejects, then both get nothing.

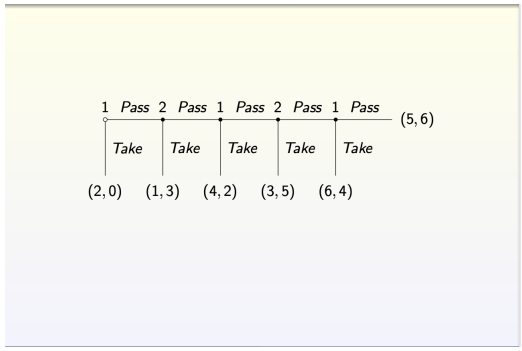
We can rewrite using our notation as

$$\begin{aligned} N &= \{1, 2\} \\ A &= \{0, 1, 2, 3, 4, a, r\} \\ Z &= \{0a, 1a, 2a, 3a, 4a, 0r, 1r, 2r, 3r, 4r\} \\ b \in \{0, 1, 2, 3, 4\} &\Rightarrow u(ba) = (4 - b, b), u(br) = (0, 0) \\ P(\emptyset) = 1, P(0) &= P(1) = P(2) = P(3) = P(4) = 2 \end{aligned}$$

Strategy for player is to choose among $\{0, 1, 2, 3, 4\}$ and the strategy from player 2 is the function that maps $S_2 : \{0, 1, 2, 3, 4\} \to \{a, r\}$, describing whether or she not accepts or rejects any given offer.

4 PIGS Cont.

Now consider the following game called The Centipede.



How should we think about players' strategies? We can solve this game back backwards induction, and putting ourselves in the shoes of the last player to understand how they would behave and iterate from there.

4.1 Subgames

A subgame of a PIG $G = \left(N, A, H, P, \{u_i\}_{i \in N}\right)$ is a game (a PIG) that starts after a given finite history $h \in H$. Formally, the subgame $G(h)$ associated with $h = (h_1, ..., h_n) \in H$ is $G(h) = \left(N, A, H_h, P_h, \{u_i\}_{i \in N}\right)$ where

$$H_h = \{(a_1, a_2, ...,) : (h_1, ..., h_n, a_1, a_2, ...) \in H\}$$

and

$$P_h(h') = P(hh')$$

where we are basically concatenating the histories.

A strategy s of G defines a strategy s_h of $G(h)$by $s_h(h') = s(hh')$. This basically defines a strategy given prefix h and what to do now that we are at h'. A strategy profile s results in the terminal history $z_h(s)$ in the subgame defined by $h \in H$. Note that $z_h(s)$ can be counterfactual in that we want to see how players would behave as if h'' was chosen over h' even though h'' may never be chosen by some rational agent.

4.2 Backwards Induction Equilibrium (BIE)

A backward induction equilibrium (BIE) of G is a strategy profile s^* s.t. for every subgame $G(h)$ it holds that $h' \to s_i^*(hh')$ is an optimal strategy in $G(h)$, given beliefs that the rest of the players behave according to s_{-i}^* (or its restriction to $G(h)$)

4.2.1 Profitable deviations

Let s be a strategy profile. We say that s_i' is a profitable deviation from s for player i **at history** h if s_i' is a strategy for G s.t.

$$u_i(z_h(s_i', s_{-i})) > u_i(z_h(s))$$

Note that a strategy profile has no profitable deviations iff it's a BIE.

Theorem 6. *The one-stage deviation principle. Let $G = \left(N, A, H, P, \{\succsim_i\}_{i \in N}\right)$ be a finite horizon, extensive-form game with perfect information. Let s be a strategy profile that is not a BIE. There exists some history h and profitable deviation $\bar{s}_i$ for a player $i = P(h)$ in $G(h)$ s.t. $\bar{s}_i(k) = s_i(k)$ for all $k \neq h$.*

Proof. Suppose s^* is not BIE. Then some player i has a profitable deviations. Meaning $\exists \hat{s}_i$ and $h \in H_i$ s.t.

$$u_i\left(z_n\left(\hat{s}_i, s^*_{-i}\right)\right) > u_i\left(z_n\left(s^*_i, s^*_{-i}\right)\right)$$

Choose a maximal such h. Now define $\bar{s}_i$ by

$$\bar{s}_i\left(\tilde{h}\right) = \begin{cases} s^*_i\left(\tilde{h}\right) & \forall \tilde{h} \in H_i, s.t. \tilde{h} \neq h \\ \hat{s}_i\left(\tilde{h}\right) & \hat{h} = h \end{cases}$$

Will finish proof next lecture. $\square$